DNS (Domain Name System)

CCNA 200-301 Study Notes - TBN Course

What happens when you type a website name in your browser?

The full journey: DNS resolution (cache > resolver > root > TLD > authoritative), the TCP 3-way handshake, the TLS handshake for HTTPS, the HTTP request-response cycle, and connection termination.

Plus: recursive vs iterative queries, DNS records (A, AAAA, CNAME, PTR), the four types of DNS servers, the October 2025 AWS DNS outage, and interview Q&A.

Reading guide: yellow highlight = key exam term. Green box = remember-this point. Blue numbers = steps in a process.

1. What Happens When You Type www.tungabadranetworks.in in a Browser

When you type www.tungabadranetworks.in into your browser and press Enter, the system must first convert this domain name into an IP address. This process is called DNS Resolution.

Step 1: Browser & Operating System Cache Check

The browser first checks:

- Browser DNS cache
- OS DNS cache
- Local hosts file

If the IP address is already available from a previous visit, the DNS lookup ends here.

Step 2: Query Sent to the DNS Resolver

If the IP is not found locally, your PC sends a recursive DNS query to the DNS resolver configured in your network settings (for example ISP DNS, Google DNS 8.8.8.8, Cloudflare 1.1.1.1).

The recursive resolver's responsibility is to find the answer on your behalf and return it to you.

Step 3: DNS Resolver Checks Its Cache

The recursive resolver checks its own DNS cache.

If the record exists and the TTL is still valid, it returns the IP to your PC.

Step 4: Iterative Lookup Begins (If the Resolver Has No Cache)

If the resolver does not have the IP, it starts the iterative DNS lookup process.

4.1 Query to Root Name Servers

The resolver sends a query to the root name servers. There are 13 logical root servers (A-M), operated by 12 independent organizations and served from hundreds of locations worldwide using anycast. Root servers do not store actual domain IPs - they respond with the address of the TLD server responsible for the domain extension (in this case, .in).

4.2 Query to TLD Servers

The resolver then queries the .in TLD server. TLD servers store information about which authoritative name server handles each specific domain. The TLD server responds with the authoritative DNS server for tungabadranetworks.in.

4.3 Query to the Authoritative DNS Server

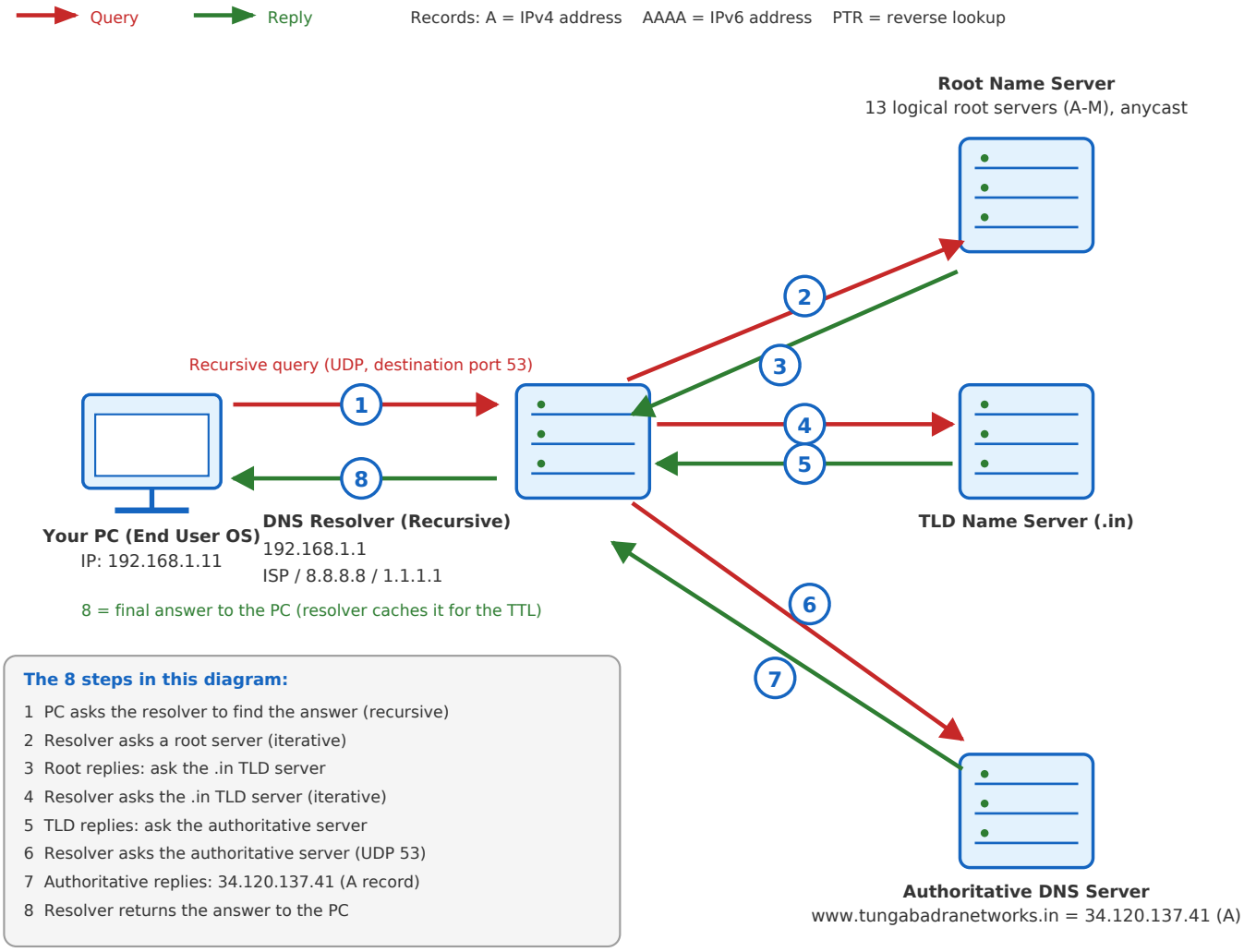
The resolver finally queries the authoritative DNS server. It returns the A record (IPv4) or AAAA record (IPv6) - the actual IP address of www.tungabadranetworks.in.

Step 5: Resolver Returns the Answer to the PC

The DNS resolver caches the IP address for the duration of the TTL, then returns the IP address to your PC through the recursive query response. DNS resolution complete - your PC now knows the IP address of the web server hosting www.tungabadranetworks.in.

The DNS Server Diagram - Full Resolution Process

This is the same lookup as a diagram, following the class example: PC 192.168.1.11 asks its resolver 192.168.1.1 for the A record of www.tungabadranetworks.in, and the resolver walks root > TLD (.in) > authoritative until it has the answer 34.120.137.41. Queries from the resolver to other servers are iterative and use UDP destination port 53.



The 8 steps in this diagram:

1. PC asks the resolver to find the answer (recursive)
2. Resolver asks a root server (iterative)
3. Root replies: ask the .in TLD server
4. Resolver asks the .in TLD server (iterative)
5. TLD replies: ask the authoritative server
6. Resolver asks the authoritative server (UDP 53)
7. Authoritative replies: 34.120.137.41 (A record)
8. Resolver returns the answer to the PC

Diagram redrawn neatly from the class notes - same devices, same 8 steps, same process.

2. TCP 3-Way Handshake (Establishing a Reliable Connection)

For classic HTTPS (HTTP/1.1 or HTTP/2), the browser first establishes a reliable connection with the server using TCP. (Newer HTTP/3 runs over QUIC/UDP instead, without TCP.)

Step 1: SYN (Synchronize) - Client to Server

The client sends a TCP packet with the SYN flag set. It also includes an Initial Sequence Number (ISN).

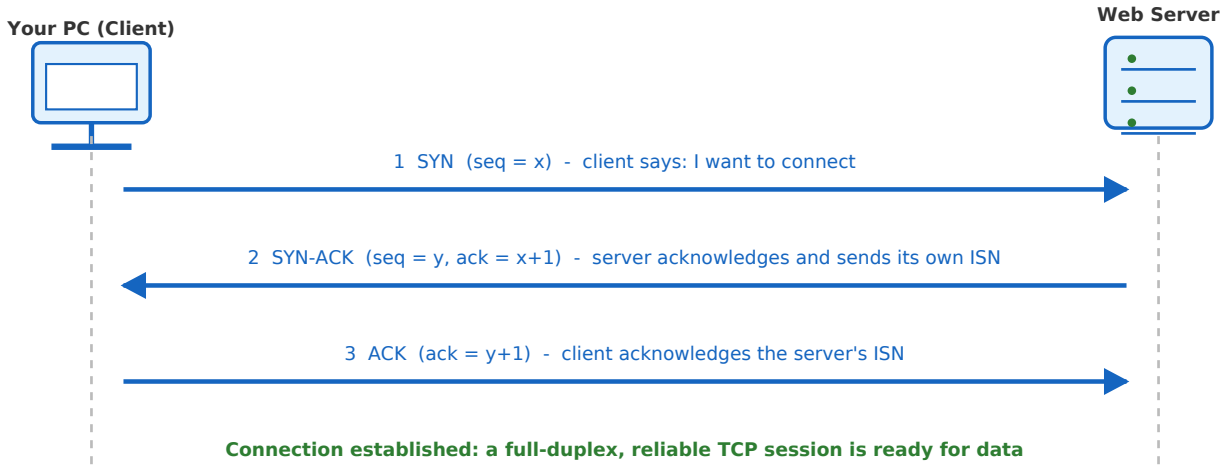
Step 2: SYN-ACK (Synchronize + Acknowledge) - Server to Client

The server responds with SYN + ACK flags. It acknowledges the client's ISN and sends its own ISN.

Step 3: ACK (Acknowledge) - Client to Server

The client sends a final ACK acknowledging the server's ISN.

Connection established - now a full-duplex, reliable TCP session is ready for data transfer.



Remember: The 3-way handshake order is SYN > SYN-ACK > ACK.

3. TLS Handshake (For Secure HTTPS Communication)

If the website uses HTTPS (secure), a TLS handshake happens after TCP to set up encryption. Simplified TLS 1.2 flow as covered in the class notes, using the classic RSA key exchange):

Step 1: ClientHello

The client sends: supported TLS versions, supported cipher suites, a random number, and session-related info.

Step 2: ServerHello + Certificate

The server responds with: the selected TLS version, the selected cipher suite, a random number, and its digital certificate (which contains the server's public key).

Step 3: Certificate Verification

The client validates the server's certificate using its trusted CA list. This ensures authenticity and security.

Step 4: Key Exchange (classic RSA method)

The client sends a pre-master secret, encrypted using the server's public key. Both sides generate: session keys, encryption keys, and MAC keys.

Step 5: ChangeCipherSpec + Finished

Both parties signal that future communication will use encrypted session keys, and encrypted "Finished" messages are exchanged.

Secure channel established - all further HTTP traffic is now encrypted and secure.

Note: modern TLS usually agrees keys with ECDHE (ephemeral Diffie-Hellman) key agreement instead of RSA key transport - both sides compute the same secret without ever sending it, which adds forward secrecy. TLS 1.3 (RFC 8446) removed RSA key exchange completely and made the handshake shorter.

4. HTTP Request-Response Cycle

After TCP and TLS are established:

Step 1: Browser Sends an HTTP Request

Example:

```
GET / HTTP/1.1
Host: www.tungabadranetworks.in
```

Step 2: Server Responds with Web Content

The server returns HTML, CSS, JavaScript, images and media.

Step 3: Browser Renders the Page

The website becomes visible to the user.

5. Connection Termination (Closing the TCP Connection)

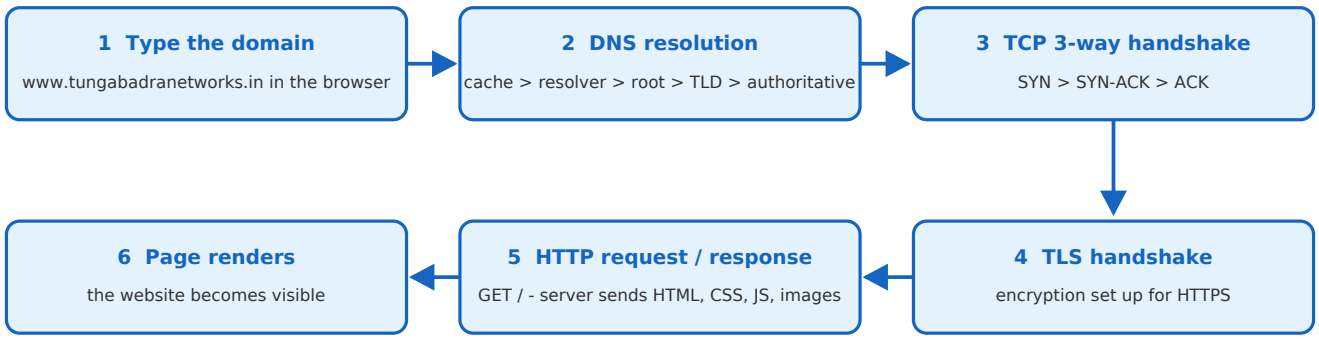
A typical graceful close uses a four-segment exchange:

- Client > Server : FIN
- Server > Client : ACK
- Server > Client : FIN
- Client > Server : ACK

After the final ACK, the connection is closed. (TCP keep-alive only probes whether an idle peer is still alive - it does not keep a connection open forever.)

Complete Flow Summary for Students

Simplified example: opening an HTTPS web page over TCP (HTTP/1.1 / HTTP/2).



- You type www.tungabadranetworks.in in the browser.
- Browser checks cache > if not found > asks the DNS resolver.
- Resolver checks its cache > if not found: queries Root > TLD (.in) > Authoritative DNS.
- Resolver returns the IP of the website.
- Browser performs the TCP 3-way handshake (SYN > SYN-ACK > ACK).
- Browser performs the TLS handshake for secure HTTPS.
- Browser sends an encrypted HTTP GET request.
- Server sends the website content.
- Browser displays the web page.

Real-World Example: DNS Outage in AWS Cloud (October 2025)

Incident Summary

On October 19-20, 2025, Amazon Web Services (AWS) had a major service disruption in its N. Virginia (us-east-1) Region. According to AWS's official post-event summary, the trigger was a defect (a race condition) in the automated DNS management system of the DynamoDB database service, which left an incorrect empty DNS record for DynamoDB's regional endpoint - so that name no longer resolved to any IP address.

Many AWS services and customer applications depend on DynamoDB, so the failure spread: new EC2 instance launches failed, some Network Load Balancers showed connection errors, and services such as Lambda, ECS/EKS and Redshift were impacted. DynamoDB's DNS records were restored within about three hours; full recovery across affected services finished the next afternoon.

Why This Matters

- A DNS failure prevented services from resolving critical internal service names - apps and APIs couldn't locate each other.
- Even though servers and data were still operational, they were unreachable by name because their network names could not be translated to IPs.
- The outage lasted hours and triggered cascading failures throughout AWS's cloud ecosystem.

Lessons from the Outage

- DNS isn't just "simple plumbing": it is a foundational service - when DNS fails, so does connectivity.
- Cascading failures: a DNS mistake inside cloud infrastructure can propagate through many dependent services.
- Resiliency planning: organizations were reminded to implement multi-region or multi-provider failover strategies.

Source: AWS official post-event summary - link in Sources and References at the end.

6. Why DNS Matters in the Real Internet

- DNS is everywhere: website access, API calls, cloud services, email routing, IoT devices.
- If DNS fails, services cannot be reached by name - even healthy servers are unreachable unless the client already has the IP cached (or connects by IP address directly).
- The AWS outage showed how even large cloud providers can be affected by a DNS issue, causing widespread disruption.

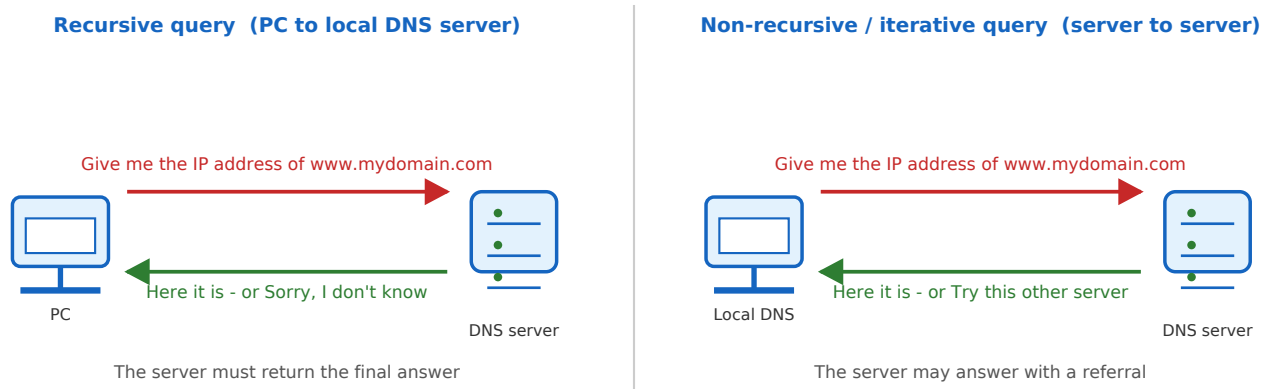
Remember: If DNS fails, names stop resolving - clients can then only reach servers whose IP addresses are already cached or known directly.

7. Recursive vs Non-Recursive (Iterative) Queries

Recursive queries are generally used by clients (PCs). They tell the DNS server to respond only with an answer, not a referral.

Non-recursive (iterative) queries are used between DNS servers. They instruct the other DNS server to return an answer, or return the address of another DNS server that may know the answer.

Iterative, in practice: the local DNS server asks the TLD server for the IP address of a given domain name. The resolver either gets the address as a response, or the IP address of another DNS server that potentially has the answer. The resolver repeats the process until the address resolution happens.



> Remember: The recursive query is between the client and its local DNS server. The iterative query is between the local DNS server and the other DNS servers (root, TLD, authoritative).

8. DNS Records

DNS records store information about domains and reside on DNS servers or in caches. The most common types:

Record	What it does
A	Maps a domain name to its IPv4 address.
AAAA	Maps a domain name to its IPv6 address.
CNAME	Maps an alias name to the canonical (real) domain name.
PTR	Maps an IP address to a domain name - used for reverse DNS lookup.

9. Types of DNS Servers

Each DNS server type plays a different role in resolving a domain name into an IP address.

1. Recursive DNS Server

A recursive DNS server (or recursive DNS resolver) resolves DNS queries by querying other DNS servers until it locates the IP address of a domain name. It acts as a mediator between the client and the other DNS servers.

2. Root DNS Server

The first point of contact from a recursive DNS server. There are 13 logical root servers, named [A-M].root-servers.net, operated by 12 independent organizations and served from hundreds of locations worldwide using anycast. Resolvers that perform iterative lookups normally start with the root hints list of all 13; forwarding resolvers can ask an upstream resolver instead. The root server provides the IP addresses of TLD servers based on the final part of the domain name. The root zone is managed by IANA (part of ICANN); the 13 root servers themselves are run by their operators.

3. TLD (Top-Level Domain) DNS Server

TLD servers resolve queries for top-level domains - for example, a .com TLD server holds delegation information for registered .com domains: which authoritative name servers serve each domain. The TLD server points the resolver to the authoritative server. IANA (a function of ICANN) maintains the root zone list of TLDs; each TLD is operated by its own registry (for example, Verisign operates .com). TLDs divide into three groups:

- Generic TLDs - common, not geographically specific: .com, .edu, .org, .gov and similar.
- Country-code TLDs - specific to a country or state: .uk, .us, .ru, .in and others.
- Infrastructure domain - .arpa (Address and Routing Parameter Area): used for reverse DNS lookups (in-addr.arpa, ip6.arpa) and other internet infrastructure.

4. Authoritative DNS Server

An authoritative DNS server provides the actual answers for DNS queries, based on the DNS records it holds for a specific domain or subdomain. It is the last step of DNS resolution. Multiple domain names can reside on one authoritative server, and a single domain name may also reside on multiple authoritative servers.

Why Do We Need Recursive Servers?

Theoretically, authoritative nameservers alone are enough to keep DNS running - but imagine the load if every complete query reached them. To decrease the load and increase speed, recursive servers (DNS resolvers) keep DNS records for a defined period of time. This time is called the TTL (time to live), and the process is named DNS caching. ISPs run such recursive servers, keeping information for many websites for the time defined by the TTL. A recursive server acts as a go-between for the customer and the authoritative DNS server - without one, every client would have to walk the root > TLD > authoritative chain itself.

Conclusion

DNS is a very useful system that saves a lot of time and makes our lives easier. After these notes you know what happens when you open a web page, how exactly DNS resolution works, and what a domain name resolution means.

Useful DNS Commands (Handy for Labs and Troubleshooting)

Command	Use
`nslookup www.tungabadranetworks.in`	Query DNS for the IP of a domain (A record lookup).
`nslookup -type=AAAA example.com`	Look up the IPv6 (AAAA) record of a domain.
`nslookup -type=PTR 34.120.137.41`	Reverse lookup - find the domain name for an IP (PTR record).
`nslookup -type=SOA example.com`	Show the zone's Start of Authority metadata (primary nameserver, admin contact, serial, timers).
`ipconfig /displaydns`	Windows: show the local DNS resolver cache.
`ipconfig /flushdns`	Windows: clear the local DNS resolver cache.
`ping www.tungabadranetworks.in`	Test name resolution plus reachability in one step.
`dig example.com`	Linux/macOS: detailed DNS query output (answer, TTL, servers used).
`host example.com`	Linux/macOS: quick name-to-IP lookup (also reverse: host 34.120.137.41).

Interview Questions and Answers

1. What is DNS (Domain Name System)?

Answer: DNS resolves a domain name into an IP address, and an IP address back into a domain name. In other words, DNS translates the domain part of the URL you type (like www.tungabadranetworks.in) into the IP address of the server that hosts it - so you never have to remember IP addresses.

2. What connection does DNS use, and what is the port number?

Answer: DNS uses port 53 for both. Normal queries and responses - including a client's recursive query - travel over UDP port 53 because they are small and fast. DNS uses TCP port 53 when reliability is needed, such as zone transfers between DNS servers or responses too large for UDP.

3. What is a DNS query and a DNS response?

Answer: A DNS query is a request sent by a client to a DNS server, asking for the IP address associated with a domain name. A DNS response is the reply from the DNS server containing the requested IP address.

4. How many root servers are there in DNS?

Answer: 13 logical root servers, named [A-M].root-servers.net, operated by 12 independent organizations and served from hundreds of locations worldwide using anycast (many physical servers stand behind those 13 names). Resolvers doing iterative lookups start from the root hints list of all 13; forwarding resolvers can ask an upstream resolver instead.

5. Which DNS record do we see for IPv4 and IPv6?

Answer: IPv4 uses the Type A record. IPv6 uses the Type AAAA record.

6. What is the difference between a recursive and an iterative query?

Answer: A recursive query (client to its local DNS server) demands the final answer - the server must not reply with a referral. An iterative query (between DNS servers) accepts either the answer or a referral to another DNS server that may know it.

7. What are the four types of DNS servers, in lookup order?

Answer: Recursive resolver (receives the client's query) > Root server (points to the right TLD) > TLD server (points to the domain's authoritative server) > Authoritative server (returns the actual record).

8. What is TTL in DNS?

Answer: Time To Live - how long a resolved record may be kept in a resolver's cache before it must be looked up again. Caching by TTL is what makes DNS fast and keeps load off the root, TLD and authoritative servers.

Sources and References

AWS official post-event summary - DynamoDB service disruption, October 19-20, 2025:
https://aws.amazon.com/message/101925/

IANA - Root Servers (the 13 identities A-M, their operators, anycast locations):
https://www.iana.org/domains/root/servers

IANA - .ARPA domain (infrastructure / reverse DNS):
https://www.iana.org/domains/arpa

RFC 1034 - Domain Names: Concepts and Facilities:
https://www.rfc-editor.org/rfc/rfc1034

RFC 1035 - Domain Names: Implementation and Specification:
https://www.rfc-editor.org/rfc/rfc1035

RFC 9293 - Transmission Control Protocol (TCP):
https://www.rfc-editor.org/rfc/rfc9293

RFC 8446 - The TLS Protocol Version 1.3:
https://www.rfc-editor.org/rfc/rfc8446

RFC 9114 - HTTP/3 (runs over QUIC/UDP instead of TCP):
https://www.rfc-editor.org/rfc/rfc9114

BIND 9 manual pages - dig, host, nslookup:
https://bind9.readthedocs.io/en/latest/manpages.html

Cloudflare Learning Center - DNS records explained:
https://www.cloudflare.com/learning/dns/dns-records/

Base source: the supplied Tungabadran Networks class notes (DNS - Domain Name System, E. Ramesh Goud). The references above ground the real-world example and the technical corrections.